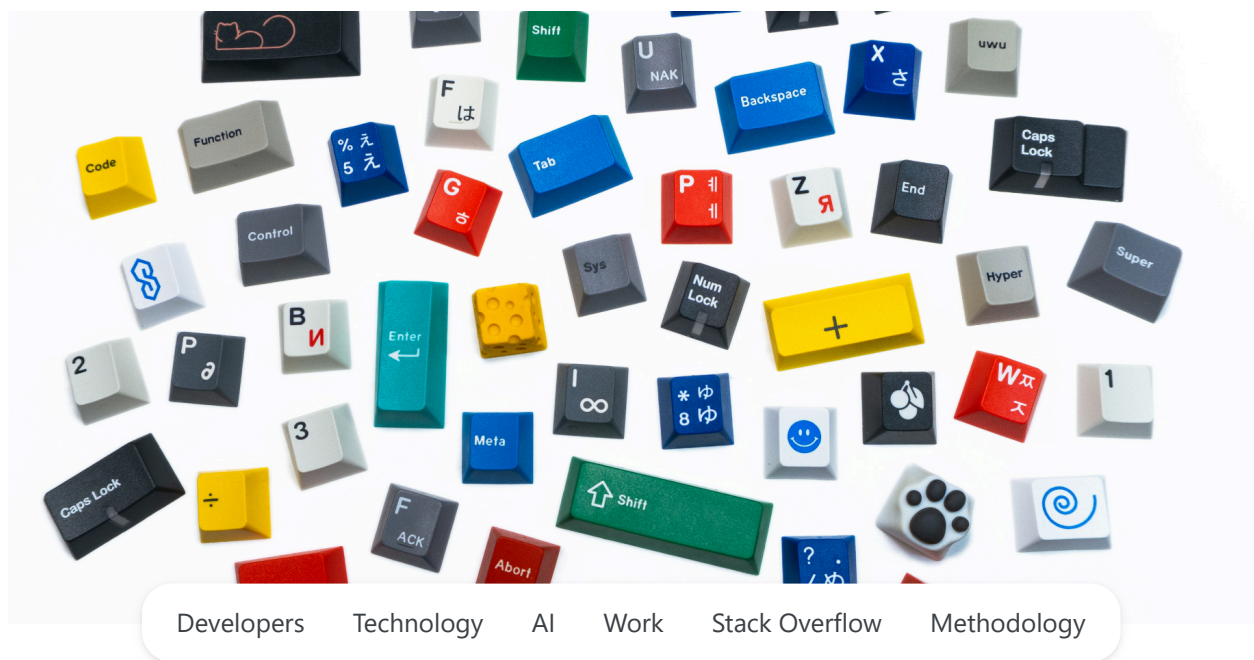




Products

The 2025 Developer Survey is the definitive report on the state of software development. In its fifteenth year, Stack Overflow received over 49,000+ responses from 177 countries across 62 questions focused on 314 different technologies, including new focus on AI agent tools, LLMs and community platforms. This annual Developer Survey provides a crucial snapshot into the needs of the global developer community, focusing on the tools and technologies they use or want to learn more about.

[Developers](#) → [Education & Experience](#)

More than one third of respondents use AI-enabled tools to learn AI this year

Yes, I learned how to use AI-enabled tools for my job or to advance my career in the last year **36.3%**

Yes, I learned how to use AI-enabled tools for my job or to advance my career in the last year **31%**

No, I learned something about AI-enabled tools for my job or to advance my career in the last year **11.7%**

No, I didn't spend time learning about AI-enabled tools for my job or to advance my career in the last year **10.8%**

No, I learned something about AI-enabled tools for my job or to advance my career in the last year **10.2%**

Over 36% of respondents learned how to use AI-enabled tools for their job or to advance their career in the last year.

[Learn to code for AI →](#)

[Developers](#) → [Education & Experience](#)

Most developers have been coding for 10+ years

1 to 5 years **13.9%**

6 to 10 years **21.1%**

11 to 15 years **15.6%**

16 to 20 years **11.8%**

21 to 30 years **14.6%**

31 to 40 years **7%**

41 to 50 years **3.1%**

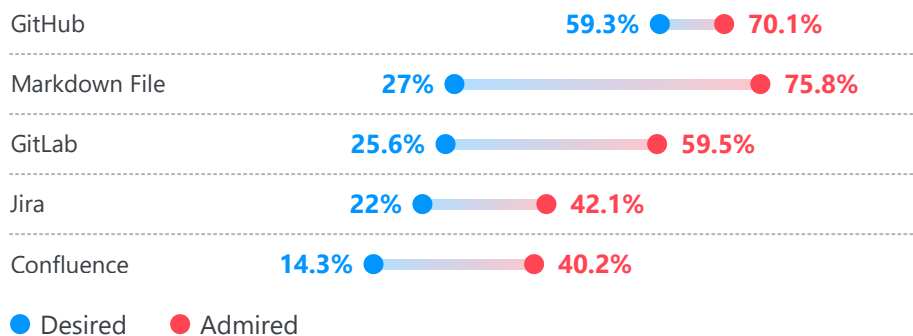
More than 50 years **0.5%**

35% of developers responding to the Developer Survey this year have been coding less than 10 years.

[Years coding →](#)

[Technology](#) → [Admired and Desired](#)

GitHub is a more desirable collaboration tool than Jira this year

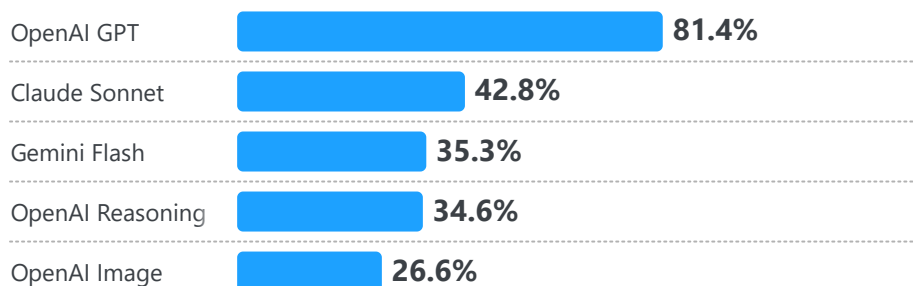


Jira steps down as the most desired tool for code documentation and collaboration and the new top desired tool is GitHub. Markdown continues to be the most admired sync tool for the third year.

[Code documentation and collaboration tools →](#)

[Technology](#) → [Most popular technologies](#)

A vast majority of developers indicating they worked with OpenAI GPT models in the past year

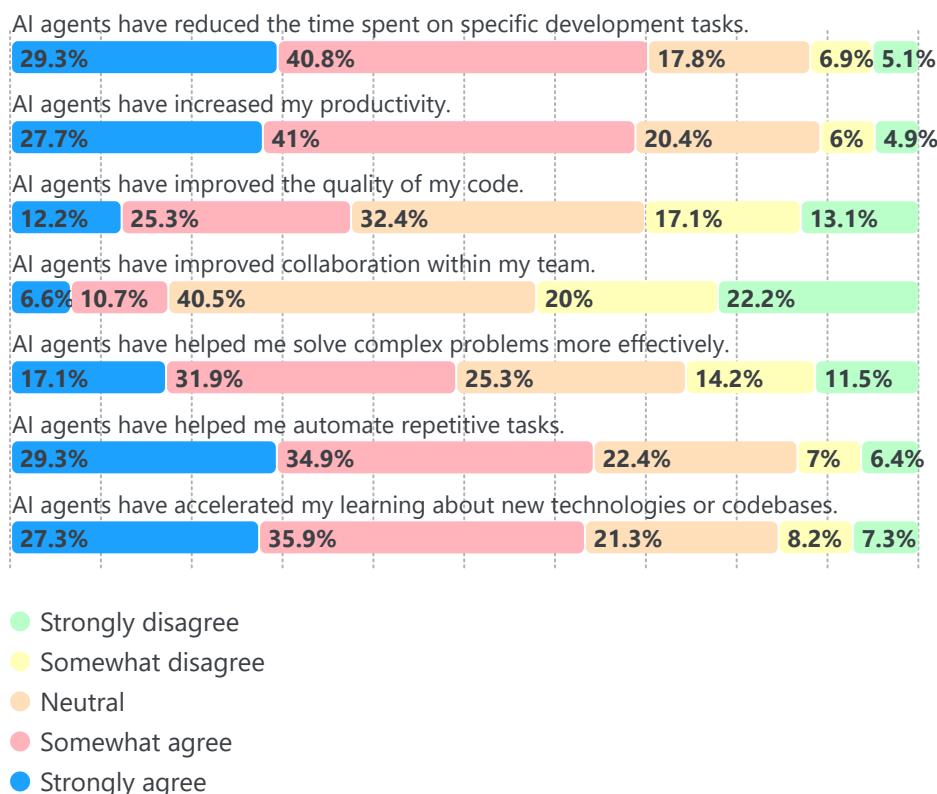


OpenAI's GPT models top the large language model list with 82% of developers indicating they used them for development work in the past year. Anthropic's Claude Sonnet models are used more by professional developers (45%) than by those learning to code (30%).

[Large language models →](#)

[AI](#) → [AI Agents](#)

69% of AI agent users agree AI agents have increased productivity



The most recognized impacts are personal efficiency gains, and not team-wide impact. Approximately 70% of agent users agree that agents have reduced the time spent on specific development tasks, and 69% agree they have increased productivity. Only 17% of users agree that agents have improved collaboration within their team, making it the lowest-rated impact by a wide margin.

[Impacts of AI agents](#) →

[AI](#) → [Sentiment and usage](#)

84% of respondents are using AI tools this year

Yes, I use AI tools daily  **47.1%**

Yes, I use AI tools weekly  **17.7%**

Yes, I use AI tools monthly  **13.7%**

No, but I plan to soon  **5.3%**

No, and I don't plan to  **16.2%**

84% of respondents are using or planning to use AI tools in their development process, an increase over last year (76%). This year we can see 51% of professional developers use AI tools daily.

[AI tools in the development process →](#)

[AI → Developer tools](#)

66% of developers are frustrated with AI solutions that are almost right

AI solutions that are almost right  **66%**

Debugging AI-generated code  **45.2%**

I don't use AI tools  **23.5%**

I've become less confident in AI  **20%**

It's hard to understand AI  **16.3%**

Other (write in):  **11.6%**

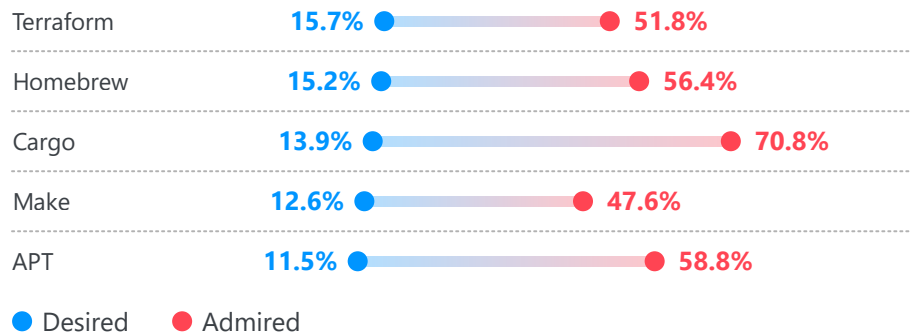
I haven't encountered any  **4%**

The biggest single frustration, cited by 66% of developers, is dealing with "AI solutions that are almost right, but not quite," which often leads to the second-biggest frustration: "Debugging AI-generated code is more time-consuming" (45%)

[AI tool frustrations →](#)

[Technology](#) → [Admired and Desired](#)

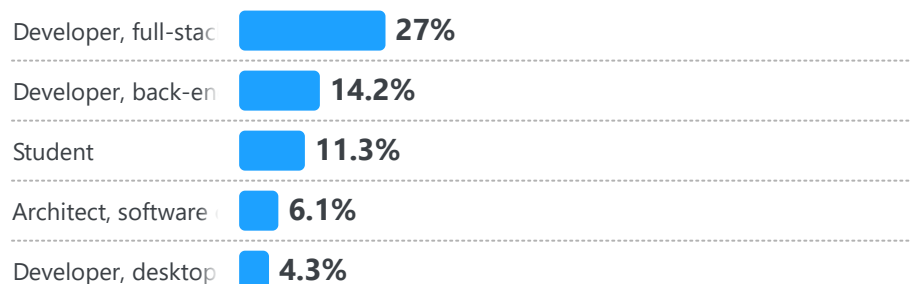
Cargo is the most admired cloud development and infrastructure tool this year



Rust's growth is directly tied to the success of its build tool and package manager, Cargo, which is the most admired (71%) cloud development and infrastructure tool this year.

[Cloud development](#) →[Developers](#) → [Developer roles](#)

Architect is the fourth top role for developers this year

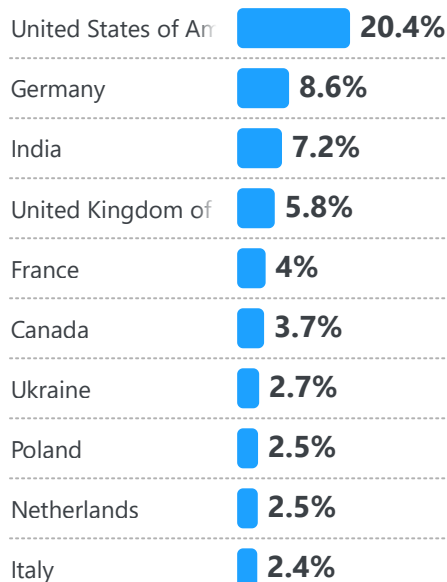


Architect is a new role we added to the survey this year and is the fourth most popular role for respondents.

[Role](#) →

[Developers](#) → [Profile](#)

USA, Germany and India are top countries responding to this year's survey

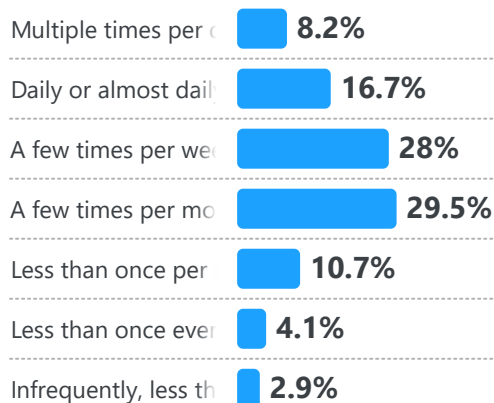


Ukraine and France swapped places this year compared to last, placing France in the top 5 list of responding countries.

Country →

[Stack Overflow](#) → [Stack Overflow site use](#)

Stack Overflow is a destination developers visit frequently

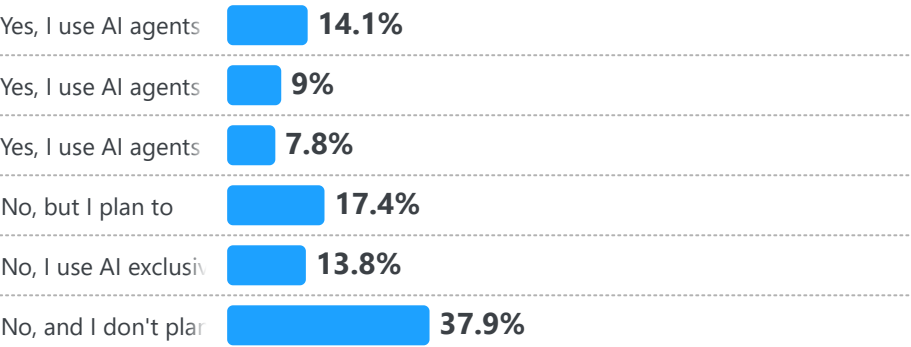


Stack Overflow is a frequent destination for information. A strong majority (82%) visit at least a few times per month, with 25% visiting daily or more often.

Frequency of visiting Stack Overflow →

AI → AI Agents

A majority of developers don't use AI agents



AI agents are not yet mainstream. A majority of developers (52%) either don't use agents or stick to simpler AI tools, and a significant portion (38%) have no plans to adopt them.

AI agents →

Work → Technology purchases

Privacy, pricing and better alternatives are top reasons developers turn their back on a technology

Detraction	Overall Rank	Median Rank	Mode Rank
Security or privacy concerns	1	3	1
Prohibitive pricing	2	4	1
Availability of better alternatives	3	4	1
Poor usability	4	4	4
Inefficient or time-costly	5	5	5
Outdated or obsolete technology or features	6	5	8

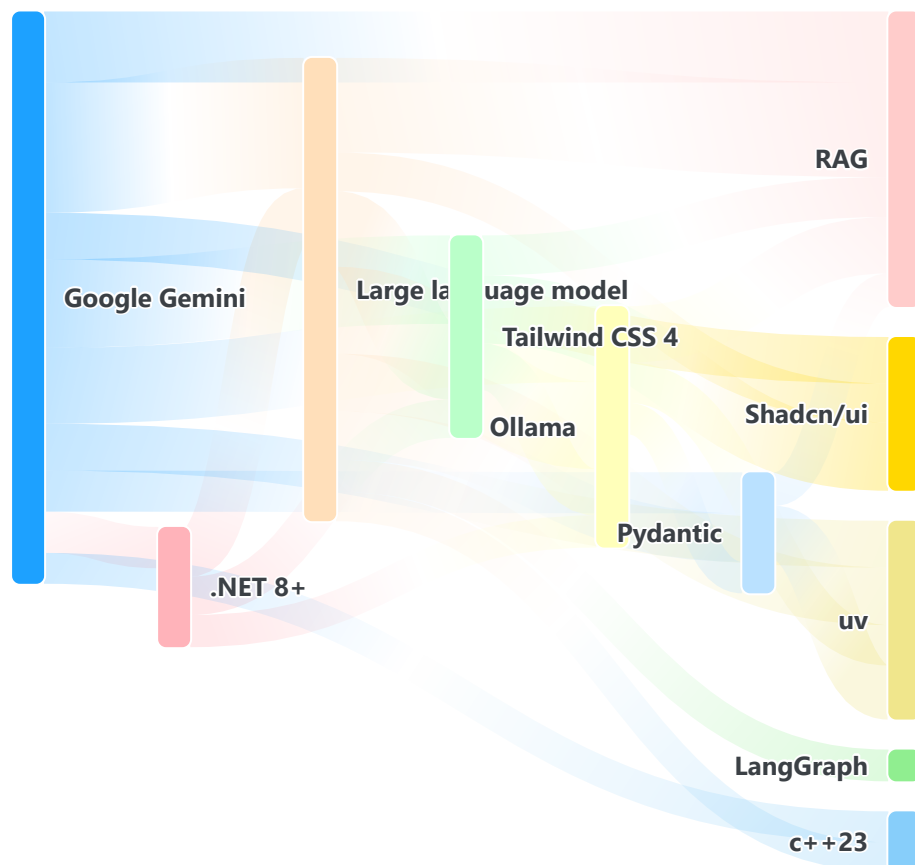
Detraction	Overall Rank	Median Rank	Mode Rank
Ethical concerns	7	6	8
Lack of or sub-par API	8	6	8
Lack of AI or AI agents	9	9	9
Other	10	10	10

The reasons to reject a technology are nearly universal. The top three deal-breakers for all developers are security or privacy concerns (Rank 1), prohibitive pricing (Rank 2), and the availability of better alternatives (Rank 3). The lack of AI is the least important factor (Rank 9).

How you lose interest in tech tools →

Technology → Worked with vs. want to work with

Developers at all levels are exploring the evolving AI landscape through Stack Overflow



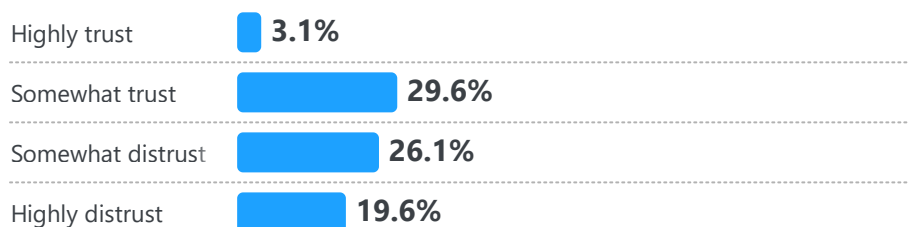
Most professional developers who indicated they used Google Gemini last year, are interested in other AI-oriented subjects like "Large Language Model" or "RAG", and tools like "Ollama".

This pattern holds true for both Professional Developers and those Learning to Code, showing that developers at all levels are actively exploring the rapidly evolving AI landscape rather than committing to a single tool or platform.

[Stack Overflow tags](#) →

[AI](#) → [Developer tools](#)

More developers actively distrust the accuracy of AI tools than trust it

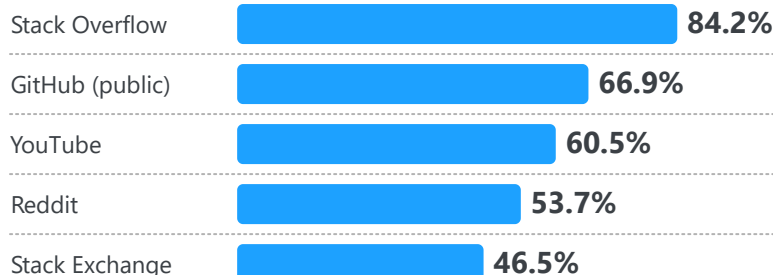


More developers actively distrust the accuracy of AI tools (46%) than trust it (33%), and only a fraction (3%) report "highly trusting" the output. Experienced developers are the most cautious, with the lowest "highly trust" rate (2.6%) and the highest "highly distrust" rate (20%), indicating a widespread need for human verification for those in roles with accountability.

[Accuracy of AI tools](#) →

[Technology](#) → [Most popular technologies](#)

Respondents learning to code use YouTube for community more than professional developers

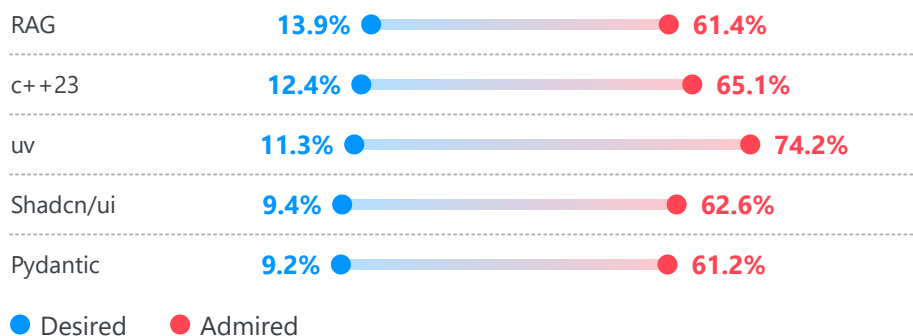


Respondents learning to code use Youtube for community more than professional developers (70% vs. 60%).

[Community platforms](#) →

[Technology](#) → [Admired and Desired](#)

uv is the most admired SO tag technology this year

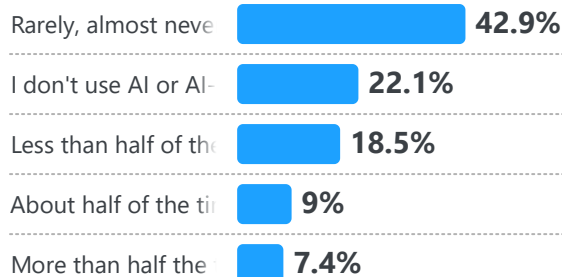


uv is a Python package manager built in Rust; need we say more about why this is the most admired (74%) SO tag technology this year?

[Stack Overflow tags](#) →

[Stack Overflow](#) → [Participation and Feedback](#)

Stack Overflow is becoming a new resource for developers that need to solve AI-related issues

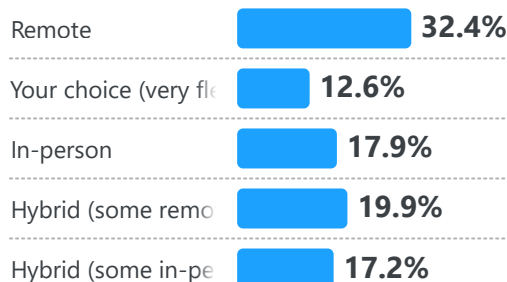


Developers turn to Stack Overflow for human-verified, trusted knowledge. About 35% of developers report that their visits to Stack Overflow are a result of AI-related issues at least some of the time.

[Frictions on Stack Overflow →](#)

[Work → Employment](#)

Nearly one third of developers are working remote this year

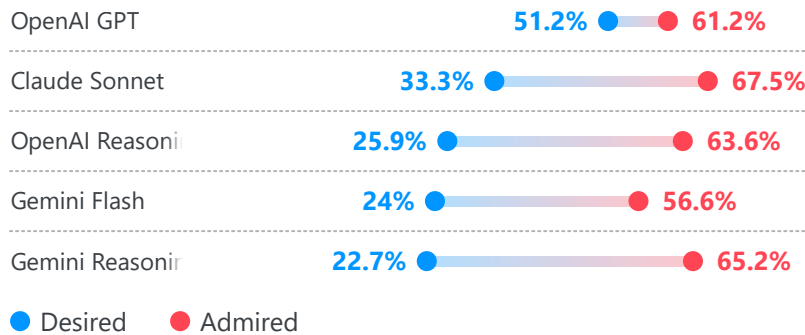


Of the top-reporting countries in this year's survey, the US has the highest number of developers working remotely (45%). 21% of developers in Germany say the choice to go into the office or work remotely is completely up to them.

[Work environment →](#)

[Technology → Admired and Desired](#)

Claude Sonnet is the most admired AI model



Anthropic's Claude Sonnet is the most admired LLM this year (behind Gemini Reasoning) and second most desired (33%).

[Large language models →](#)

[Work → Job Satisfaction](#)

One in four developers are happy at their current job

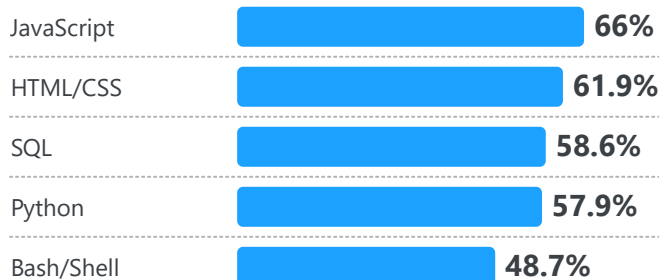


More developers are happy at work this year (24% vs. 20% last year). This is likely related to that pay bump in the data for certain roles this year.

[Job satisfaction →](#)

[Technology → Most popular technologies](#)

Python adoption grew in 2025

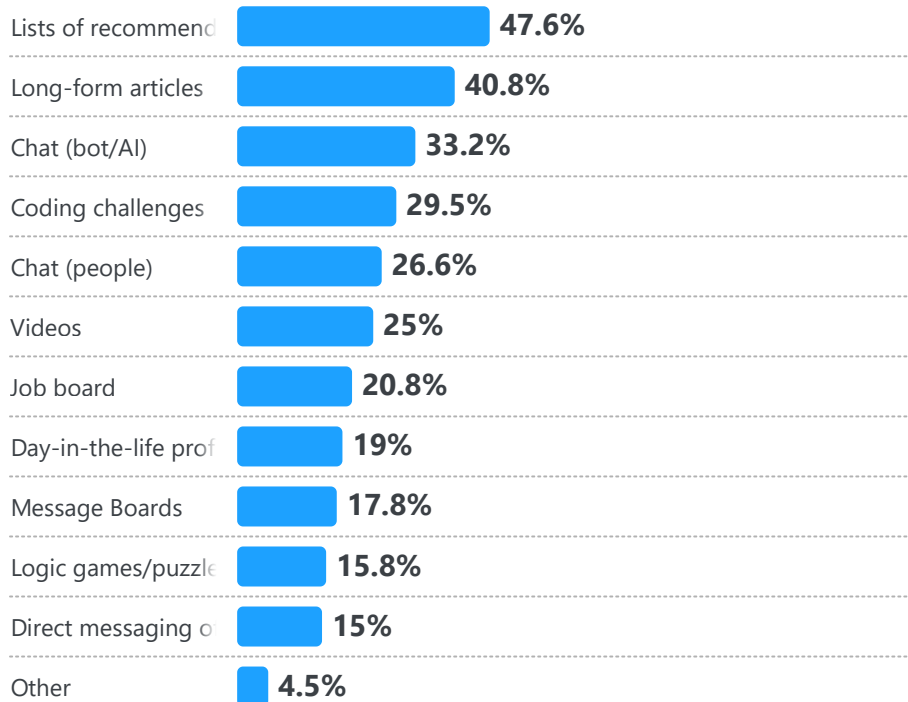


After more than a decade of steady growth, Python's adoption has accelerated significantly. It saw a 7 percentage point increase from 2024 to 2025; this speaks to its ability to be the go-to language for AI, data science, and back-end development.

Programming, scripting, and markup languages →

[Stack Overflow](#) → [Participation and Feedback](#)

Younger developers want developer content with social or interactive formats



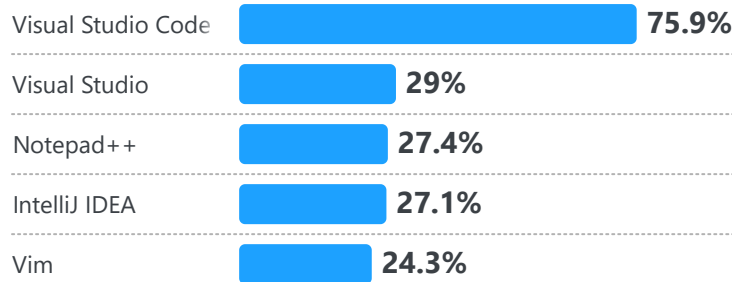
While all age groups want lists and articles, younger developers show a significantly higher interest in more social and interactive formats. For example, 37% of 18-24 year olds want "Chat (people)", compared to only 20% of 55-64 year olds. Similarly, 39% of the youngest cohort want "Coding challenges," also aligning with a motivation to skill up.

How do you choose to find relevant developer content? →

[Technology](#) → [Most popular technologies](#)

Visual Studio and Visual Studio Code both maintained their top spots for developer

environments used for the fourth year

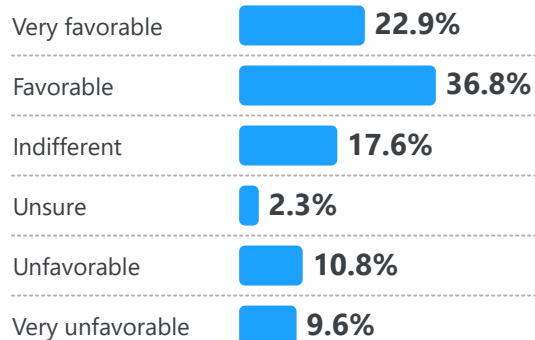


Subscription-based, AI-enabled IDEs weren't able to topple the dominance of Visual Studio and Visual Studio Code this year. Both maintained their top spots for the fourth year while relying on extensions as optional, paid AI services.

[Dev IDEs →](#)

[AI → Sentiment and usage](#)

Positive sentiment to AI tools has decreased in 2025



Conversely to usage, positive sentiment for AI tools has decreased in 2025: 70%+ in 2023 and 2024 to just 60% this year. Professionals show a higher overall favorable sentiment (61%) than those learning to code (53%).

[AI tool sentiment →](#)

Survey index

1. Developers

2. Technology

3. AI

4. Work

5. Stack Overflow

1.1. Profile Coding Age Country	2.1. Most popular technologies Programming, scripting, and markup languages Databases Cloud development Web frameworks and technologies Dev IDEs Stack Overflow tags Community platforms Large language models Code documentation and collaboration tools Computer operating systems	3.1. Sentiment and usage AI tools in the development process AI tool sentiment	4.1. Employment Employment status Employment additional status Work environment IC or PM	5.1. Stack Overflow use Have a Stack Overflow account How long have you had a Stack Overflow account Frequency of Stack Overflow use
1.2. Education & Experience Educational attainment Years coding Learn to code How did you learn to code? Learn to code for AI How did you learn to code for AI?		3.2. Developer tools Accuracy of AI tools AI tools' ability to handle complex tasks AI in the development workflow AI workflow and tool satisfaction AI tool frustrations AI and humans in the future Vibe coding	4.2. Company info Company size	5.2. Participation and impact Frequency of participation on Stack Overflow How do you find developers on Stack Overflow How do you use Stack Overflow Feel like a Stack Overflow community member Friction on Stack Overflow
1.3. Developer roles Role Work experience Industry	2.2. Admired and Desired Programming, scripting, and markup languages Databases Cloud development Web frameworks and technologies Dev IDEs Stack Overflow tags	3.3. AI Agents AI agents AI agents affect on work productivity AI agent uses at work AI agent uses for general purposes	4.3. Salary Salary by developer type Salary and experience by developer type	4.4. Technology purchases Influence on technology purchases What makes you a fan of specific tech tools How you lose interest in tech tools Count of tools used to work Count of tools used to code outside of work 4.5. Job Satisfaction

Community platforms	Impacts of AI agents	Looking for a new role
Large language models	Challenges with AI agents	Job satisfaction
Code documentation and collaboration tools	AI Agent data storage tools	Ranked attributes contributing to or detracting from job satisfaction
2.3. Worked with vs. want to work with Programming, scripting, and markup languages Databases Cloud development Web frameworks and technologies Dev IDEs Stack Overflow tags Community platforms Large language models Code documentation and collaboration tools	AI Agent orchestration tools	Is AI a threat to your job?
	AI Agent observability and security	
	AI Agent out-of-the-box tools	



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